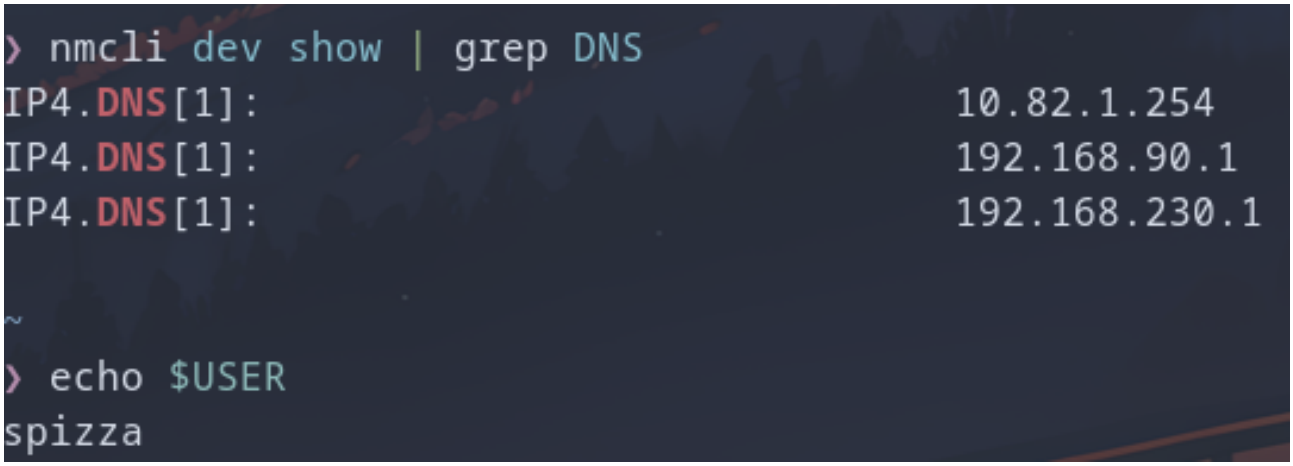
DNS Command-Line Tools

Protokoll — NSCS 4. Klasse · 26.05.2026 · Prof. Fuxreiter

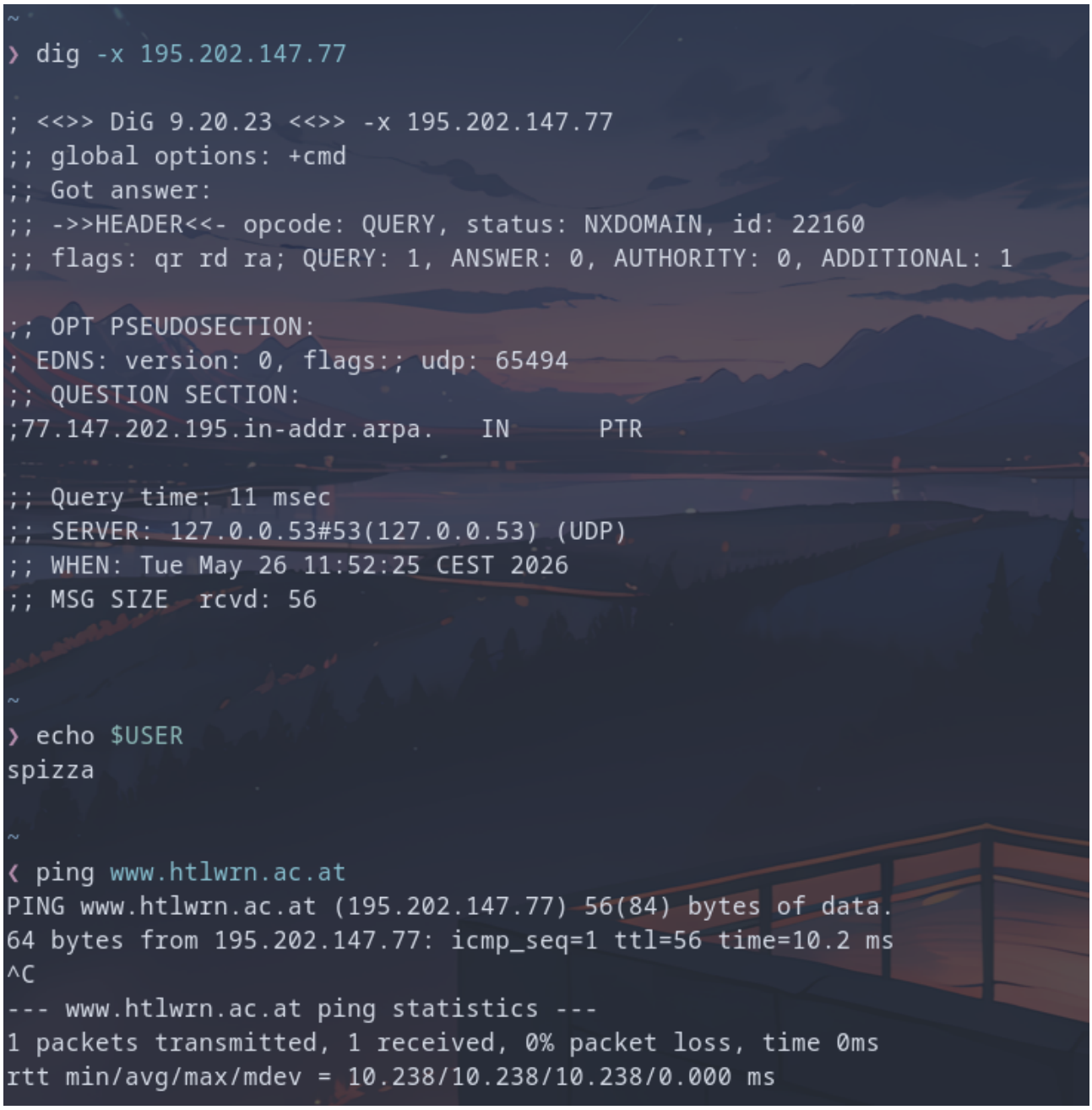
Aufgabe 1: Welche DNS-Server verwendet mein PC?

IP-Adresse der DNS-Server (resolvectl dns zeigt den tatsächlichen Server hinter 127.0.0.53).



Aufgabe 2: IP-Adresse von www.htlwrn.ac.at

Forward-Lookup und Überprüfung eines Reverse-DNS-Eintrags.



Aufgabe 3: IP-Adresse von dns.google

Forward-Lookup, Reverse-DNS-Eintrag prüfen und Forward/Reverse vergleichen.



```
❯ ping dns.google
PING dns.google (8.8.8.8) 56(84) bytes of data.
64 bytes from dns.google (8.8.8.8): icmp_seq=1 ttl=116 time=16.6 ms
^C
--- dns.google ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 16.639/16.639/16.639/0.000 ms

~
❯ dig -x 8.8.8.8

; <<>> DiG 9.20.23 <<>> -x 8.8.8.8
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 10918
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 3

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;8.8.8.8.in-addr.arpa.          IN      PTR

;; ANSWER SECTION:
8.8.8.8.in-addr.arpa.   163     IN      PTR     dns.google.

;; ADDITIONAL SECTION:
dns.google.             163     IN      A       8.8.8.8
dns.google.             163     IN      A       8.8.4.4

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue May 26 11:56:53 CEST 2026
;; MSG SIZE  rcvd: 105
```

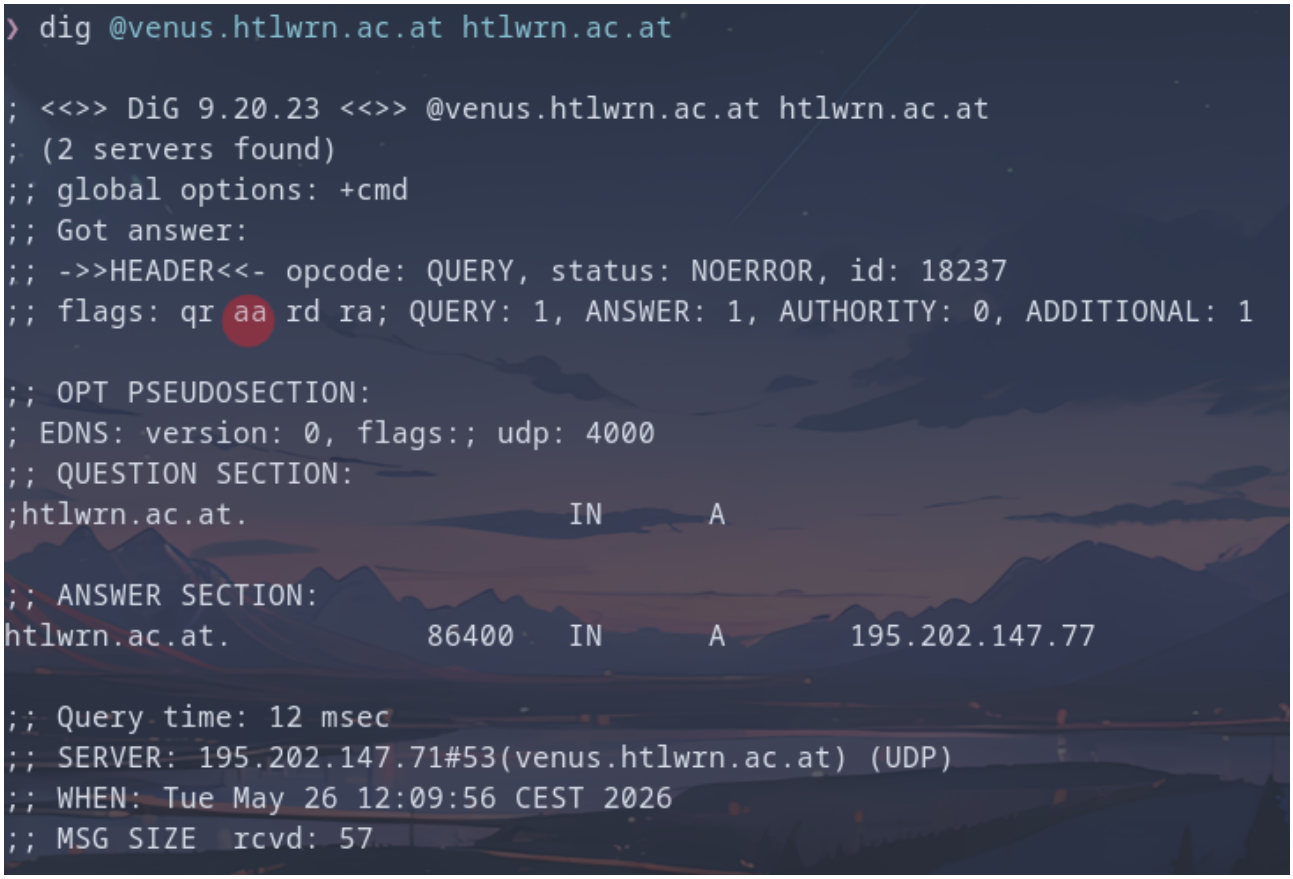
Aufgabe 4: Autoritative Nameserver für orf.at und google.com

NS-Records für beide Domains ermitteln.



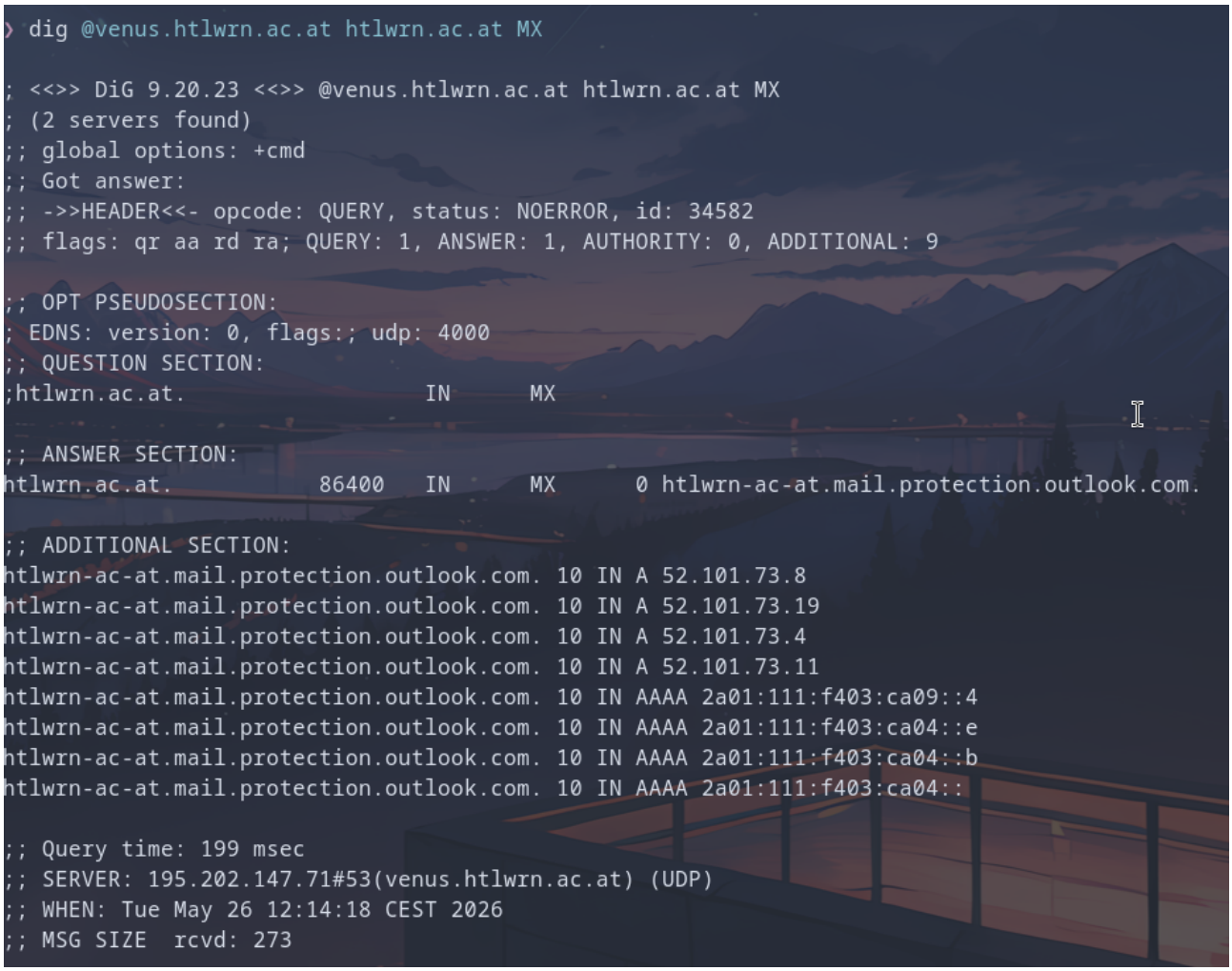
Aufgabe 5: IP-Adressen direkt vom autoritativen Nameserver

www.google.com und www.instagram.com — autoritative Antwort erkennbar am Fehlen des 'Non-authoritative answer'-Hinweises.



Aufgabe 6: Mailserver (MX-Records) für orf.at und gmail.com

Name und IP-Adresse der zuständigen Mailserver.



```
❯ dig @venus.htlwrn.ac.at htlwrn.ac.at MX

; <<>> DiG 9.20.23 <<>> @venus.htlwrn.ac.at htlwrn.ac.at MX
; (2 servers found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 34582
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 9

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;htlwrn.ac.at.                  IN      MX

;; ANSWER SECTION:
htlwrn.ac.at.           86400   IN      MX      0 htlwrn-ac-at.mail.protection.outlook.com.

;; ADDITIONAL SECTION:
htlwrn-ac-at.mail.protection.outlook.com. 10 IN A 52.101.73.8
htlwrn-ac-at.mail.protection.outlook.com. 10 IN A 52.101.73.19
htlwrn-ac-at.mail.protection.outlook.com. 10 IN A 52.101.73.4
htlwrn-ac-at.mail.protection.outlook.com. 10 IN A 52.101.73.11
htlwrn-ac-at.mail.protection.outlook.com. 10 IN AAAA 2a01:111:f403:ca09::4
htlwrn-ac-at.mail.protection.outlook.com. 10 IN AAAA 2a01:111:f403:ca04::e
htlwrn-ac-at.mail.protection.outlook.com. 10 IN AAAA 2a01:111:f403:ca04::b
htlwrn-ac-at.mail.protection.outlook.com. 10 IN AAAA 2a01:111:f403:ca04::

;; Query time: 199 msec
;; SERVER: 195.202.147.71#53(venus.htlwrn.ac.at) (UDP)
;; WHEN: Tue May 26 12:14:18 CEST 2026
;; MSG SIZE  rcvd: 273
```

Aufgabe 7: CNAMEs / Aliase für news.orf.at und www.apple.com

CNAME-Kette bis zum finalen A-Record auflösen.

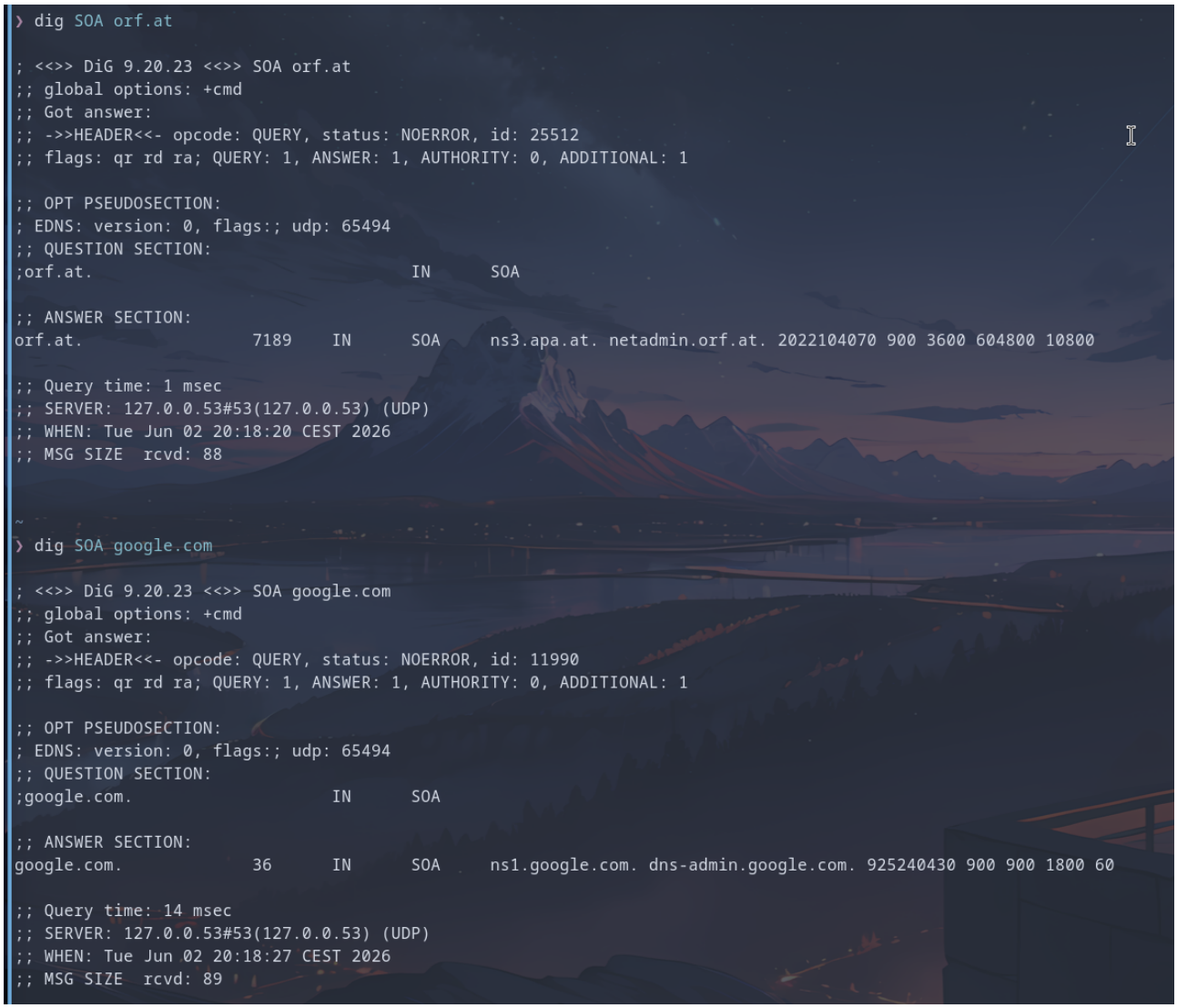


```
❯ dig news.orf.at CNAME +short
www.orf.at.

~
❯ dig www.apple.com CNAME +short
www-apple-com.v.aaplimg.com.
```

Aufgabe 8: SOA-Metadaten für orf.at und google.com

Seriennummer, Refresh, Retry, Expire, Minimum-TTL aus dem Start-of-Authority-Record.



```
❯ dig SOA orf.at

; <<>> DiG 9.20.23 <<>> SOA orf.at
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25512
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;orf.at.                                IN      SOA

;; ANSWER SECTION:
orf.at.                 7189    IN      SOA     ns3.apa.at. netadmin.orf.at. 2022104070 900 3600 604800 10800

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Jun 02 20:18:20 CEST 2026
;; MSG SIZE  rcvd: 88


~
❯ dig SOA google.com

; <<>> DiG 9.20.23 <<>> SOA google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 11990
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      SOA

;; ANSWER SECTION:
google.com.             36      IN      SOA     ns1.google.com. dns-admin.google.com. 925240430 900 900 1800 60

;; Query time: 14 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Jun 02 20:18:27 CEST 2026
;; MSG SIZE  rcvd: 89
```

Aufgabe 9: Default-TTL-Vergleich: google.com vs. orf.at

google sind 60 sekunden, orf 10800

Aufgabe 10: E-Mail-Adressen der Zone-Admins (google.com und orf.at)

orf: netadmin.orf.at => netadmin@orf.at
google: dns-admin.google.com => dns-admin@google.com
